INCH POUND

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1 July 1991

DETAIL SPECIFICATION

# ALUMINUM ALLOY ARMOR ROLLED PLATE (1/2 TO 4 INCHES THICK), WELDABLE (ALLOY 2519)

This specification is approved for use within the Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers aluminum alloy 2519 armor rolled plate from 1/2 inch up to and including 4 inches in thickness.

1.2 Weldability. The material covered by this specification is weldable to itself and other weldable 2000 series alloys using a 2000 series filler and is weldable to 6000 series alloys using a 4000 series filler. It is not weldable to 5000 series or 7000 series alloys by conventional welding techniques.

1.3 Classification. Material furnished under this specification should be rolled plate in a T87 temper

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified in the contract or purchase order, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Research Laboratory, Weapons and Materials Research Directorate, ATTN: AMSRL-WM-M, Aberdeen Proving Ground, MD 21005-5069 by using the Standardization Document Improvement Proposal (DD from 1426) appearing at the end of this document or by letter.



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STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-662 Test Method Standard $V_{50}$ Ballistic Test for Armor

(Unless otherwise indicated, copies of the standard are available from the Standardization Document Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified in the contract or purchase order, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified in the contract or purchase order, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

SAE INTERNATIONAL

AMS 2750 Pyrometry

AMS 2772 Heat Treatment of Aluminum Alloy Raw Materials

(Application for copies should be addressed to SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI H35.2 American National Standard Dimensional Tolerances for Aluminum Mill Products

(Applications for copies should be addressed to the American National Standards Institute, Customer Service, 11 West 42nd St., New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B557 Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products

B666 Identification Marking of Aluminum Products

E34 Methods for Chemical Analysis of Aluminum and Aluminum Alloys

E227 Method for Optical Emission Spectrometric Analysis of Aluminum and Aluminum Alloys by the Point-to-Plane Technique

E527 Practice for Numbering Metals and Alloys (UNS)

E607 Method for Optical Emmission Spectrometric Analysis of Aluminum and Aluminum Alloys by the Point-to-Plane Technique, Nitrogen Atmosphere

E716 Practices for Sampling Aluminum and its Alloys for Spectrochemical Analysis



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1251 Optical Emission Spectrometric Analysis of Aluminum and Aluminum Alloys by Argon Atmosphere, Point to Plane, Unipolar Self-initiating Capacitor Discharge Chemical Analysis

G47 Determining Susceptibility to Stress-Corrosion Cracking of High Strength Aluminum Alloy Products

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2) and before production has commenced, samples of the specified item shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples shall be manufactured by the process proposed for use on production.

3.2 Chemical composition. Chemical composition of material furnished under this specification shall be within the limits specified in table I (see 6.2).

3.3 Mechanical properties. Unless otherwise specified in the contract or order (see 6.2), the mechanical properties of rolled plate shall meet the requirements of table II, when tested as specified in 4.7.2.

3.3.1 If mechanical property requirements differ from the values shown in table II, or if any other properties are required, the ballistic requirements shall be as agreed upon between the procuring activity and the manufacturer.

3.4 Ballistic requirements. Each ballistic test plate shall be proof fired for resistance to penetration by determining a protection ballistic limit, BL(P) (see 6.2). The BL(P) shall be as specified in appendix A.



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TABLE I. Chemical composition, percent (ladle analysis).

| Element | Min. | Max. |
|---|---|---|
| Silicon | - | 0.25 1/ |
| Iron | - | 0.30 1/ |
| Copper | 5.3 | 6.4 |
| Manganese | 0.10 | 0.50 |
| Magnesium | 0.05 | 0.40 |
| Titanium | 0.02 | 0.10 |
| Zinc | - | 0.10 |
| Zirconium | 0.10 | 0.25 |
| Vanadium | 0.05 | 0.15 |
| Others, each | - | 0.05 |
| Others, total | - | 0.15 |
| Aluminum | Remainder | |

1/ The combined silicon and iron content shall not exceed 0.40 percent.

3.5 Stress-corrosion resistance. When stress corrosion testing is specified in the contract or order (see 6.2), plate 0.75 inch and greater in thickness shall be resistant to stress corrosion cracking, with 3 out of 3 specimens not showing evidence of cracking when tested in accordance with 4.7.3.

3.6 Thermal processing. Heat treatment shall conform to the requirements of AMS 2750 and 2772 and shall be such as to enable the material to meet the requirements of these specifications.

3.7 Dimensions and tolerances.

3.7.1 Dimensions. Size of plate shall be as specified in the invitation for bids, contract or order (see 6.2).



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TABLE II. Mechanical properties of plate.

| Direction | Thickness, inches | Tensile strength min, ksi | Yield strength 0.2% offset min, ksi | Elongation in 2 inches min, percent |
|---|---|---|---|---|
| Long transverse | 0.500-1.500 | 68.0 | 58.0 | 7 |
| | 1.501-2.000 | 68.0 | 58.0 | 7 |
| | 2.001-3.000 | 68.0 | 59.0 | 6 |
| | 3.001-4.000 | 68.0 | 59.0 | 5 |
| Longitudinal | 0.500-1.500 | 66.0 | 59.0 | 10 |
| | 1.501-2.000 | 66.0 | 59.0 | 9 |
| | 2.001-3.000 | 67.0 | 60.0 | 8 |
| | 3.001-4.000 | 68.0 | 61.0 | 7 |

3.7.2 Tolerances. Unless otherwise specified in the contract or order (see 6.2), the plates submitted for acceptance shall not vary from the specified dimensions by an amount greater than the tolerances shown in ANSI H35.2. Similarly, the thickness tolerance shall be as specified in table III.

TABLE III. Thickness tolerances.

| Ordered thickness, inches | Tolerance, inch Minus | Plus |
|---|---|---|
| 0.500-0.875, incl. | 0.030 | 1/ |
| 0.876-1.125, incl. | 0.035 | -- |
| 1.126-1.375, incl. | 0.040 | -- |
| 1.376-1.625, incl. | 0.045 | -- |
| 1.626-1.875, incl. | 0.050 | -- |
| 1.876-2.250, incl. | 0.055 | -- |
| 2.251-2.750, incl. | 0.060 | -- |
| 2.751-3.000, incl. | 0.065 | -- |
| 3.001-4.000, incl. | 0.070 | -- |

1/ The values for the plus column are derived from the full range tolerances specified in table 3.1 (for sheet and plate not specified for aerospace application) of ANSI H35.2. For example, the tolerance for a 2.0 inch thick by a 73 inches wide plate would be as follows:

| | |
|---|---|
| Thickness tolerance (from ANSI H35.2) | ± 0.060 |
| Full tolerance less minus tolerance | 0.120 - 0.055 = 0.065 |
| Derived plus tolerance | 0.065 |
| Tabulated minus tolerance | 0.055 |



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3.8 Marking for identification. Each plate shall be marked in at least one location in characters not less that 3/8 inch high with the following information.

a. Manufacturer's name, trademark, or both.
b. Number of this specification.
c. Type, class, alloy designation, and temper.
d. Army part or drawing number.
e. Lot number.

3.8.1 Identification marking. Unless otherwise specified (see 6.2) each plate shall be continuously line marked along the rolling direction in accordance with ASTM B666.

3.8.2 Ballistic test plates. In addition to the marking in 3.8, each ballistic test plate shall be marked with the letters PRE for first article test plates and ACC for acceptance test plates. This marking shall be impression stamped or stenciled on the plate rolled surface not less than 1/2 inch high and within one inch of the plate edge.

3.9 Workmanship. Rolled plate shall be uniform in quality and shall be clean, smooth and commercially free from buckles, blisters, hard spots, damaged ends, laminations and other defects which may affect its use in fabrication. Surface cracks, edge cracks or edge laminations shall be cause for rejection.

4. VERIFICATION

4.1 Clarification of inspection. The inspection requirements specified herein are classified as conformance inspection (see 4.2).

4.2. Conformance inspection. Test for acceptance of individual lots consist of testing all requirements specified in section 3. Verification conformance inspection is in accordance with the provisions herein.

4.2.1 Government test activity. Ballistic tests shall be performed at a government test activity selected by the procuring activity (see 6.2).

4.3. Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (see 4.3.1).
(b) Quality conformance inspection (see 4.3.2).

4.3.1 First article inspection. First article inspection shall consist of the following:

(a) Chemical analysis (see 3.2 and 4.7.1).
(b) Mechanical properties (see 3.3 and 4.7.2).
(c) Ballistic tests (see 3.4, 4.7.4 and Appendix A).
(d) Stress corrosion tests (see 3.5, 4.5.2.3 and 4.7.3).
(e) Dimensions and tolerances (see 3.7 and 4.8.1).
(f) Identification marking (see 3.8 and 4.8.2).
(g) Workmanship (see 3.9 and 4.8.2).



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4.3.2 Conformance (acceptance) inspection. Lot acceptance inspection shall consist of the following:

(a) Chemical analysis (see 3.2 and 4.7.1).
(b) Mechanical properties (see 3.3 and 4.7.2).
(c) Ballistic tests (see 3.4, 4.7.4 and Appendix A).
(d) Dimensions and tolerances (see 3.7 and 4.8.1).
(e) Identification marking (see 3.8 and 4.8.2).
(f) Workmanship (see 3.9 and 4.8.2).
(g) Preparation for shipment (see section 5 and 4.8.3).

4.4 Lot. For rolled plate a lot shall consist of plate of the same ordered thickness and temper submitted for inspection as a unit, provided that the weight of the finished plate in the lot does not exceed 50,000 pounds.

4.5 Sampling.

4.5.1 First article testing. At least three samples shall be submitted for ballistic and mechanical testing. If the sample size is too small for removal of tensile test specimens from the ballistic sample, three additional samples shall be furnished.

4.5.1.1 Chemical composition. Samples for chemical analysis shall be removed from each plate selected for ballistic testing.

4.5.1.2 Mechanical properties. A tensile test specimen for first article tests shall be removed from the same plate material that has been selected for the ballistic test. The location and type specimen shall be specified in 4.7.2.

4.5.1.3 Stress corrosion. Stress corrosion test specimens shall be removed from the same plate selected for the ballistic test. The type of specimens shall be as specified in 4.7.3.

4.5.1.4 Ballistic tests. Two plates, 12 by 36 inches (+1,- 0 inches) by the ordered thickness, shall be submitted for ballistic testing as agreed between the manufacturing and the procuring activity. The orientation of these plates with respect to the rolling direction shall be at the option of the manufacturer. The first article ballistic test may be waived at the discretion of the procuring activity, provided that the manufacturer has produced acceptable plate within $\pm$ 0.250 inch of the ordered thickness within 18 months and that the test conditions are the same as for previously accepted plate.

4.5.2 Acceptance tests.

4.5.2.1 Chemical composition.

(a) Ladle analysis. At least one sample shall have been taken from the molten metal representing one group of ingots poured as a unit from the same source of molten metal. Complete ingot analysis records shall be available to the Government at the ingot producer's facility.

(b) Check analysis. When sampling has not been made in accordance with 4.5.2.1(a), one sample shall be taken for each 4,000 pounds or less in each lot in accordance with methods specified in 4.7.1.



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4.5.2.2 Mechanical properties. From each lot samples for tensile tests shall be selected in accordance with table IV. Each sample shall be selected from a different plate in the lot. If a lot consists of only one plate, only one tensile test sample shall be taken. One tensile specimen shall be fabricated from each sample.

4.5.2.3 Stress corrosion. One plate, approximately 12 inches by 12 inches by the ordered thickness shall be selected from the lot for stress corrosion tests. At the discretion of the procuring activity, reduced sampling may be instituted, i.e., not every lot need be sampled (see 4.9.2.3.2).

TABLE IV. Number of tensile test samples.

| Lot size, pounds | Minimum number of samples |
|---|---|
| To 8,000, incl. | 2 |
| 8,001-12,000, incl. | 3 |
| 12,001-20,000, incl | 4 |
| Over 20,000, incl. | 5 |

4.5.2.4 Ballistic testing. One plate approximately 12 inches by 36 inches by the ordered thickness shall be selected from the lot for ballistic testing. The orientation of this plate with respect to the rolling direction shall be at the option of the manufacturer.

4.6 Test specimens.

4.6.1 Chemical analysis. Specimens for chemical analysis shall be prepared in accordance with the applicable method specified in 4.7.1.

4.6.2 Mechanical properties. Tension test specimens shall conform to the dimensions of the round 0.50 inch diameter or proportional subsized standard specimens of ASTM B557. They shall be taken in the long transverse direction. From plate in nominal thickness 0.50 to 1.50 inches inclusive, tensile test specimens shall be taken with the axis midway between the two plate surfaces; from plate in nominal thicknesses greater than 1.50 inches, the axis of the tensile test specimen shall be three-fourths of the distance from one surface to the other.

4.6.3 Stress corrosion. The stress corrosion specimens shall be stressed at 75% of the specified yield strength. The location of the specimens taken from the plate shall be specified in the drawing (see 6.2).

4.7 Test procedures.

4.7.1 Chemical composition. The determination of the chemical composition shall be made in accordance with one or more of the following test methods: ASTM E34, E716, E227, E607, and E1251.



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4.7.2 Mechanical properties. The mechanical properties shall be determined in accordance with ASTM B557.

4.7.3 Resistance to stress corrosion. Stress corrosion testing shall be conducted in accordance with ASTM G47.

4.7.4 Ballistic tests: Plates shall be tested using projectiles chosen in accordance with the thickness ranges shown in table V. Plate thickness as determined by the ballistic test agency shall be used to determine the required ballistic limit for that plate. Individual thickness measurements are to be read from a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 inch and the average of these readings reported to the nearest 0.005 inch. Measurements shall be made on the intended impact area.

4.8 Examination.

4.8.1 Dimensions. Each plate within the lot shall be measured to determine compliance with the requirements for dimensions and tolerances (see 3.7) or the lot shall be sampled in accordance with sampling procedures approved by the procuring activity.

4.8.2 Visual. Each plate shall be examined to determine compliance with the identification marking (see 3.8) and workmanship (see 3.9) requirements.

TABLE V. Acceptance ballistic test plates.

| Ordered or nominal design thickness specified on drawings, inches | Test angle | Projectile to be used |
|---|---|---|
| 0.500-0.749 | $30^{o}$ | Cal. .30 AP, M2 |
| 0.750-1.500 | $0^{o}$ | Cal. .30 AP, M2 |
| 0.750-0.950 | $0^{o}$ | Cal. .50 FS |
| 0.951-1.500 | $0^{o}$ | 20 mm FS |
| (a) 1.501-3.000 | $0^{o}$ | Cal. .50 AP, M2 |
| (a) 3.001-4.000 | $0^{o}$ | 14.5 mm API, BS41 |

(a) Ballistic limit requirements have not been developed for this thickness range.

4.8.3 Preparation for shipment. Prior to shipment, examination shall be made to determine compliance with section 5.

4.9 Rejection and retest.

4.9.1 First article. Failure of the first article test plates to meet the requirements of 4.3.1 indicates rejection of the product and process.



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4.9.1.1 Retests. Resubmission of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfaction of the procuring activity. Samples submitted for retest shall be marked R1, R2 and R3, if applicable, in addition to the requirements of 3.8.

4.9.2 Acceptance.

4.9.2.1 Examination. A quality conformance lot shall be rejected for failure to meet the visual, dimensional and preparation for delivery requirements when examined in accordance with 4.8.

4.9.2.2 Tests. A quality conformance lot shall be rejected for failure to meet any of the requirements of any of the tests outlined in 4.3.2.

4.9.2.3 Retests.

4.9.2.3.1 Chemical composition and mechanical properties. Retests shall be conducted in accordance with the methods specified in 4.7.1 and 4.7.2.

4.9.2.3.2 Stress corrosion (first article) testing only. If the manufacturer so desires, a retest shall be performed using three additional specimens. No cracking shall occur in 10 days. If the retest also fails, the manufacturer may elect to resubmit the lot after retreatment of the entire lot. After retreatment, the lot must conform to the requirements outlined in 3.5.

4.9.2.3.3 Ballistic. Rejection and retest of ballistic test plates shall be in accordance with A.5.2.

4.9.2.4 Reduced testing. At the discretion of the procuring activity, the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product, meeting the testing requirements is being produced and provided the manufacturer agrees to maintain the same manufacturing procedures.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The material specified herein is for military vehicles.



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6.1.1 Title. The title of the specification has been changed from "Aluminum Alloy Armor Rolled Plate (1/4 to 4 Inches Thick), Weldable (Alloy 2519)" to "Aluminum Alloy Armor Rolled Plate (1/2 to 4 Inches Thick), Weldable (Alloy 2519)”

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) If first article inspection is required (see 3.1).
(c) If stress corrosion testing is required (see 3.5).
(d) Drawings, dimensions and tolerances (see 3.7, table III)
(e) If different identification marking is required (see 3.8).
(f) Destination of ballistic test samples (see 4.5.1.4).

6.3 Definitions.

6.3.1 Manufacturer. The manufacturer is defined as the company producing the aluminum alloy plate.

6.3.2 Contractor. The contractor is the company which has a direct contract from the Government to furnish an end item -- usually a vehicle. The contractor is also known as the prime contractor.

6.3.3 Contracting officer. The term "contracting officer" means the person executing a contract on behalf of the Government and any other officer or civilian employee who is properly designated contracting officer; and the term includes, except as otherwise provided, the authorized representative of the contracting officer acting within the limits of that person’s authority.

6.3.4 Procuring activity. The term "procuring activity" is that activity of the Government which actually initiates the request for procurement and maintains the records of the procurement.

6.4 Mechanical properties and ballistic requirements. The minimum mechanical properties specified (see 3.3) may not assure aluminum armor material meeting the specified ballistic requirements (see 3.4).

6.5 Metric units. When metric dimensions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

| To go from English | Multiply by | To get Metric SI Unit |
|---|---|---|
| inch | 0.0254 | meter (m) |
| foot | 0.3048 | meter (m) |
| foot pound force | l.3558 | joule (J) |
| feet per second | 0.3048 | meter per second (m/s) |
| pounds/per square inch | 0.00689 | megapascal (MPa) |



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6.6 Patent notice. The material specified by this specification may be covered by U.S. Patent # 4610733 (expiration date September 9, 2003.) The Government does not have a royalty free license for this patent.

6.7 Key words.

Armor plate
Aluminum alloy 2519

Custodian:
Army - MR

Review Activities:
Army - AT, AR, TE
DLA - IS

Preparing activity:
Army - MR

Project 9535-A046



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APPENDIX A

BALLISTIC TESTING OF ALUMINUM ALLOY ARMOR PLATE

A.1 SCOPE. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor, plate, weldable, when tested in accordance with the provisions of this specification. This Appendix is a mandatory part of this specification.

A.2 APPLICABLE DOCUMENTS

A.2.1 The following document forms a part of this appendix (see A.5.1):

MIL-STD-662, Test Method Standard $V_{50}$ Ballistic Test for Armor

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

A.3 DEFINITIONS

A.3.1 Fair impact. A fair impact results when an unyawed projectile strikes an unsupported area of the ballistic test plate sample at such a location that there is at least two calibers of undisturbed metal between this impact and any previous impact, hole, crack, edge of plate or spalled area.

A.3.2 Witness plate. A witness plate is normally a 0.014 inch thick sheet of 5052 H36 aluminum alloy (or a 0.020 inch thick sheet of 2024-T3 aluminum alloy placed 6 inches behind and parallel to the test plates or other ballistic sample.

A.3.3 Protection complete penetration, CP(P). A protection complete penetration will have been obtained when the projectile or a fragment of the projectile or a fragment of the ballistic test plate is thrown beyond the rear face of the plate with sufficient energy to perforate an approximate 0.020 inch thick sheet of 2024-T3 aluminum alloy (or a 0.014 inch thick sheet of 5052 H36) placed parallel to and 6 inches behind the plate.

A.3.4 Partial penetration, PP(P). Any fair impact that is not a complete penetration should be considered a partial penetration.

A.3.5 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

A.4 REQUIREMENTS



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A.4.1 Resistance to penetration. The minimum required $V_{50}$ ballistic limit shall be in accordance with the values shown in tables A-I through A-IV.

A.5 TESTS

A.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, Test Method Standard $V_{50}$ Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.5.1.1 Temperature conditioning. Prior to the test, the test item(s) will be temperature conditioned at least eight hours. Thermostatic control will be such that the average temperature of the item during the test shall be $72^{o} \pm 15^{o}F$ ($22^{o} \pm 8^{o}C$).

A.5.1.2 Protection ballistic limit, BL(P).

A.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) will be reported.

A.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data will be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit will be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing will continue until a 10-round ballistic limit has been attained using the smallest practical velocity spread. Ten round ballistic limits will be reported as such on the armor data forms.

A.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit which has been attained is within $\pm$ 10 fps of the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, will be fired to reduce the gap to 25 fps or less. The ballistic limit will then be recomputed using the above criteria. The recomputed BL(P) will be reported as the BL(P) of the plate. In borderline cases, a reduction of the gap between the high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P).



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A.5.2 Rejection and retest of ballistic plates.

A.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

A.5.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material as acknowledged by the procuring activity. Two retest plates shall be submitted for first article testing and both tests must pass; otherwise, the armor material shall be rejected.

A.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or order, failure of a test plate to meet the ballistic requirements indicates rejection of the lot; however, the final decision will depend on the outcome of retests, if submitted.

A.5.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure, may submit at his own expense two additional test plates from the same lot for ballistic retest. If either of these plates fail the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fail, the lot shall be permanently rejected.

A.5.3 Disposition of ballistic test plates.

A.5.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates will be returned "as is" to the applicant, at his expense, unless the plates were destroyed in testing.

A.5.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered as part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered as part of the lot they represent and remain the property of the producer just as the rejectable lot does. The failed plates will be returned, upon request, as in A.5.3.1.



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APPENDIX A

TABLE A-I. Minimum required ballistic limits -.30 caliber AP M2 projectile at 30 degrees obliquity - 2519 alloy.

| Thickness (inches) | Required BL(P) fps | Thickness (inches) | Required BL(P) fps | Thickness (inches) | Required BL(P) fps |
|---|---|---|---|---|---|
| 0.470 | l382 | 0.595 | l683 | 0.720 | l939 |
| 0.475 | 1395 | 0.600 | 1694 | 0.725 | 1949 |
| 0.480 | 1409 | 0.605 | 1705 | 0.730 | 1958 |
| 0.485 | 1422 | 0.6l0 | 1716 | 0.735 | 1968 |
| 0.490 | 1434 | 0.6l5 | 1726 | 0.740 | 1977 |
| 0.495 | 1447 | 0.620 | 1737 | 0.745 | 1987 |
| *0.500 | 1460 | 0.625 | 1748 | **0.749 | 1994 |
| 0.505 | 1472 | 0.630 | 1758 | 0.750 | 1996 |
| 0.510 | 1485 | 0.635 | 1769 | 0.755 | 2005 |
| 0.515 | 1497 | 0.640 | 1779 | 0.760 | 2015 |
| 0.520 | 1509 | 0.645 | 1790 | 0.765 | 2024 |
| 0.525 | 1522 | 0.650 | 1800 | 0.770 | 2033 |
| 0.530 | 1534 | 0.655 | 1810 | 0.775 | 2042 |
| 0.535 | 1546 | 0.660 | 1821 | 0.780 | 2051 |
| 0.540 | 1557 | 0.665 | 1831 | 0.785 | 2060 |
| 0.545 | 1569 | 0.670 | 1841 | 0.790 | 2069 |
| 0.550 | 1581 | 0.675 | 1851 | 0.795 | 2078 |
| 0.555 | 1593 | 0.680 | 1861 | 0.800 | 2087 |
| 0.560 | 1604 | 0.685 | 1871 | 0.805 | 2096 |
| 0.565 | 1616 | 0.690 | 1881 | 0.8l0 | 2105 |
| 0.570 | 1627 | 0.695 | 1891 | 0.815 | 2114 |
| 0.575 | 1638 | 0.700 | 1900 | 0.820 | 2123 |
| 0.580 | l650 | 0.705 | l910 | 0.825 | 2l32 |
| 0.585 | l661 | 0.7l0 | l920 | 0.830 | 2l40 |
| 0.590 | l672 | 0.7l5 | l930 | | |

* Specification requirements begin for ordered thickness 0.500 inch.
** Specification requirements end for ordered thickness 0.749 inch.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.



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APPENDIX A

TABLE A-II. Minimum required ballistic limits -.30 caliber AP M2 projectile at 0 degrees obliquity - 2519 alloy.

| Thickness (inches) | Required BL(P) fps | Thickness (inches) | Required BL(P) fps | Thickness (inches) | Required BL(P) fps |
|---|---|---|---|---|---|
| 0.720 | 1770 | 0.920 | 2062 | 1.120 | 2317 |
| 0.725 | 1778 | 0.925 | 2068 | 1.125 | 2323 |
| 0.730 | 1786 | 0.930 | 2075 | 1.130 | 2329 |
| 0.735 | 1794 | 0.935 | 2082 | 1.135 | 2335 |
| 0.740 | 1802 | 0.940 | 2088 | 1.140 | 2341 |
| 0.745 | 1809 | 0.945 | 2095 | 1.145 | 2347 |
| * 0.750 | 1817 | 0.950 | 2102 | 1.150 | 2353 |
| 0.755 | 1825 | 0.955 | 2l08 | 1.155 | 2359 |
| 0.760 | 1832 | 0.960 | 2115 | 1.160 | 2365 |
| 0.765 | 1840 | 0.965 | 2122 | 1.165 | 2371 |
| 0.770 | 1847 | 0.970 | 2128 | 1.170 | 2377 |
| 0.775 | 1855 | 0.975 | 2135 | 1.175 | 2382 |
| 0.780 | 1862 | 0.980 | 2141 | 1.180 | 2388 |
| 0.785 | 1870 | 0.985 | 2148 | 1.l85 | 2394 |
| 0.790 | 1877 | 0.990 | 2154 | 1.l90 | 2400 |
| 0.795 | 1885 | 0.995 | 2161 | 1.l95 | 2406 |
| 0.800 | 1892 | 1.000 | 2167 | 1.200 | 2412 |
| 0.805 | 1900 | 1.005 | 2174 | 1.205 | 2417 |
| 0.810 | 1907 | 1.0l0 | 2180 | 1.210 | 2423 |
| 0.815 | 1914 | 1.015 | 2187 | 1.215 | 2429 |
| 0.820 | l921 | l.020 | 2193 | l.220 | 2435 |
| 0.825 | l929 | l.025 | 2199 | l.225 | 2440 |
| 0.830 | l936 | l.030 | 2206 | l.230 | 2446 |
| 0.835 | l943 | l.035 | 22l2 | l.235 | 2452 |
| 0.840 | l950 | l.040 | 2218 | l.240 | 2458 |
| 0.845 | l957 | l.045 | 2225 | l.245 | 2463 |
| 0.850 | l964 | l.050 | 2231 | l.250 | 2469 |
| 0.855 | l972 | l.055 | 2237 | l.255 | 2475 |
| 0.860 | l979 | l.060 | 2243 | l.260 | 2480 |
| 0.865 | l986 | l.065 | 2250 | l.265 | 2486 |
| 0.870 | 1993 | l.070 | 2256 | l.270 | 2492 |
| 0.875 | 2000 | l.075 | 2262 | l.275 | 2497 |
| 0.880 | 2007 | l.080 | 2268 | l.280 | 2503 |
| 0.885 | 2014 | l.085 | 2274 | l.285 | 2508 |
| 0.890 | 2021 | l.090 | 2280 | l.290 | 2514 |
| 0.895 | 2027 | l.095 | 2287 | l.295 | 2520 |
| 0.900 | 2034 | l.l00 | 2293 | l.300 | 2525 |
| 0.905 | 2041 | l.l05 | 2299 | l.305 | 2531 |
| 0.9l0 | 2048 | l.ll0 | 2305 | l.3l0 | 2536 |
| 0.9l5 | 2055 | l.ll5 | 23l1 | l.3l5 | 2542 |



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APPENDIX A

TABLE A-II. Minimum required ballistic limits -.30 caliber AP M2 projectile at 0 degrees obliquity - 2519 alloy (continued).

| Thickness (inches) | Required BL(P) fps | Thickness (inches) | Required BL(P) fps | Thickness (inches) | Required BL(P) fps |
|---|---|---|---|---|---|
| 1.320 | 2547 | 1.425 | 2660 | 1.530 | 2769 |
| 1.325 | 2553 | 1.430 | 2665 | 1.535 | 2774 |
| 1.330 | 2558 | 1.435 | 2671 | 1.540 | 2779 |
| 1.335 | 2564 | 1.440 | 2676 | 1.545 | 2784 |
| 1.340 | 2569 | 1.445 | 2681 | 1.550 | 2789 |
| 1.345 | 2575 | 1.450 | 2686 | 1.555 | 2794 |
| 1.350 | 2580 | 1.455 | 2692 | 1.560 | 2799 |
| 1.355 | 2585 | 1.460 | 2697 | 1.565 | 2804 |
| 1.360 | 2591 | 1.465 | 2702 | 1.570 | 2809 |
| 1.365 | 2596 | 1.470 | 2707 | 1.575 | 2814 |
| 1.370 | 2602 | 1.475 | 2712 | 1.580 | 2819 |
| 1.375 | 2607 | 1.480 | 2718 | 1.585 | 2824 |
| 1.380 | 2612 | 1.485 | 2723 | 1.590 | 2829 |
| 1.385 | 2618 | 1.490 | 2728 | 1.595 | 2834 |
| 1.390 | 2623 | 1.495 | 2733 | 1.600 | 2839 |
| 1.395 | 2628 | ** 1.500 | 2738 | 1.605 | 2844 |
| 1.400 | 2634 | 1.505 | 2743 | 1.610 | 2849 |
| 1.405 | 2639 | 1.510 | 2748 | 1.615 | 2854 |
| 1.410 | 2644 | 1.515 | 2753 | 1.620 | 2858 |
| 1.415 | 2650 | 1.520 | 2759 | 1.625 | 2863 |
| 1.420 | 2655 | 1.525 | 2764 | | |

* Specification requirements begin for ordered thickness 0.750 inch.
**Specification requirements end for ordered thickness 1.500 inch.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.



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APPENDIX A

TABLE A-III. Minimum required ballistic limits .50 caliber fragment simulating projectile at 0 degrees obliquity.

| Thickness (inches) | Required BL(P) fps | Thickness (inches) | Required BL(P) fps | Thickness (inches) | Required BL(P) fps |
|---|---|---|---|---|---|
| 0.720 | 1827 | 0.825 | 2274 | 0.930 | 2823 |
| 0.725 | 1847 | 0.830 | 2297 | 0.935 | 2852 |
| 0.730 | 1866 | 0.835 | 2321 | 0.940 | 2881 |
| 0.735 | 1886 | 0.840 | 2345 | 0.945 | 2911 |
| 0.740 | 1905 | 0.845 | 2370 | ** 0.950 | 2941 |
| 0.745 | 1925 | 0.850 | 2394 | 0.955 | 2971 |
| * 0.750 | 1946 | 0.855 | 2419 | 0.960 | 3002 |
| 0.755 | 1966 | 0.860 | 2444 | 0.965 | 3033 |
| 0.760 | 1987 | 0.865 | 2469 | 0.970 | 3064 |
| 0.765 | 2007 | 0.870 | 2495 | 0.975 | 3095 |
| 0.770 | 2028 | 0.875 | 2521 | 0.980 | 3127 |
| 0.775 | 2050 | 0.880 | 2547 | 0.985 | 3159 |
| 0.780 | 2071 | 0.885 | 2573 | 0.990 | 3192 |
| 0.785 | 2093 | 0.890 | 2600 | 0.995 | 3225 |
| 0.790 | 2114 | 0.895 | 2627 | 1.000 | 3258 |
| 0.795 | 2136 | 0.900 | 2654 | 1.005 | 3291 |
| 0.800 | 2159 | 0.905 | 2681 | 1.010 | 3325 |
| 0.805 | 2181 | 0.910 | 2709 | 1.015 | 3359 |
| 0.810 | 2204 | 0.915 | 2737 | 1.020 | 3394 |
| 0.815 | 2227 | 0.920 | 2765 | 1.025 | 3428 |
| 0.820 | 2250 | 0.925 | 2794 | 1.030 | 3462 |

* Specification requirements begin for this ordered thickness 0.750 inch.
** Specification requirements end for ordered thickness 0.950 inch.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.



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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - 20 mm fragment simulating projectile at 0 degrees obliquity – 2519 alloy.

| Thickness (inches) | Required BL(P) fps | Thickness (inches) | Required BL(P) fps | Thickness (inches) | Required BL(P) fps |
|---|---|---|---|---|---|
| 0.915 | 1269 | 1.110 | 1663 | 1.310 | 2184 |
| 0.920 | 1278 | 1.115 | 1675 | 1.315 | 2199 |
| 0.925 | 1287 | 1.120 | 1686 | 1.320 | 2214 |
| 0.930 | 1296 | 1.125 | 1698 | 1.325 | 2229 |
| 0.935 | 1305 | 1.130 | 1710 | 1.330 | 2244 |
| 0.940 | 1314 | 1.135 | 1721 | 1.335 | 2259 |
| 0.945 | 1324 | 1.140 | 1733 | 1.340 | 2274 |
| 0.950 | 1333 | 1.145 | 1745 | 1.345 | 2290 |
| * 0.951 | 1335 | 1.150 | 1757 | 1.350 | 2305 |
| 0.955 | 1342 | 1.155 | 1769 | 1.355 | 2321 |
| 0.960 | 1352 | 1.160 | 1781 | 1.360 | 2336 |
| 0.965 | 1361 | 1.165 | 1793 | 1.365 | 2352 |
| 0.970 | 1371 | 1.170 | 1806 | 1.370 | 2368 |
| 0.975 | 1380 | 1.175 | 1818 | 1.375 | 2384 |
| 0.980 | 1390 | 1.180 | 1831 | 1.380 | 2400 |
| 0.985 | 1400 | 1.185 | 1843 | 1.385 | 2416 |
| 0.990 | 1409 | 1.190 | 1856 | 1.390 | 2433 |
| 0.995 | 1419 | 1.195 | 1868 | 1.395 | 2449 |
| 1.000 | 1429 | 1.200 | 1881 | 1.400 | 2465 |
| 1.005 | 1439 | 1.205 | 1894 | 1.405 | 2482 |
| 1.010 | 1449 | 1.210 | 1907 | 1.410 | 2499 |
| 1.015 | 1459 | 1.215 | 1920 | 1.415 | 2516 |
| 1.020 | 1469 | 1.220 | 1933 | 1.420 | 2532 |
| 1.025 | 1479 | 1.225 | 1946 | 1.425 | 2550 |
| 1.030 | 1490 | 1.230 | 1960 | 1.430 | 2567 |
| 1.035 | 1500 | 1.235 | 1973 | 1.435 | 2584 |
| 1.040 | 1510 | 1.240 | 1986 | 1.440 | 2601 |
| 1.045 | 1521 | 1.245 | 2000 | 1.445 | 2619 |
| 1.050 | 1531 | 1.250 | 2014 | 1.450 | 2636 |
| 1.055 | 1542 | 1.255 | 2027 | 1.455 | 2654 |
| 1.060 | 1553 | 1.260 | 2041 | 1.460 | 2672 |
| 1.065 | 1563 | 1.265 | 2055 | 1.465 | 2690 |
| 1.070 | 1574 | 1.270 | 2069 | 1.470 | 2708 |
| 1.075 | 1585 | 1.275 | 2083 | 1.475 | 2726 |
| 1.080 | 1596 | 1.280 | 2097 | 1.480 | 2744 |
| 1.085 | 1607 | 1.285 | 2111 | 1.485 | 2763 |
| 1.090 | 1618 | 1.290 | 2126 | 1.490 | 2781 |
| 1.095 | 1629 | 1.295 | 2140 | 1.495 | 2800 |
| 1.100 | 1641 | 1.300 | 2155 | ** 1.500 | 2819 |
| 1.105 | 1652 | 1.305 | 2169 | 1.505 | 2838 |



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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - 20 mm fragment simulating projectile at 0 degrees obliquity - 2519 alloy (continued).

| Thickness (inches) | Required BL(P) fps | Thickness (inches) | Required BL(P) fps | Thickness (inches) | Required BL(P) fps |
|---|---|---|---|---|---|
| 1.510 | 2857 | 1.550 | 3013 | 1.590 | 3178 |
| 1.515 | 2876 | 1.555 | 3033 | 1.595 | 3199 |
| 1.520 | 2895 | 1.560 | 3053 | 1.600 | 3220 |
| 1.525 | 2914 | 1.565 | 3074 | 1.605 | 3242 |
| 1.530 | 2934 | 1.570 | 3094 | 1.610 | 3263 |
| 1.535 | 2953 | 1.575 | 3115 | 1.615 | 3285 |
| 1.540 | 2973 | 1.580 | 3136 | 1.620 | 3307 |
| 1.545 | 2993 | 1.585 | 3157 | 1.625 | 3329 |

* Specification requirements begin for ordered thickness 0.951 inch.
** Specification requirements end for ordered thickness l.500 inches.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.



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|---|---|---|

**DOCUMENT TITLE ALUMINUM ALLOY ARMOR ROLLED PLATE (1/2 TO 4 INCHES THICK), WELDABLE (ALLOY 2519)**

**4. NATURE OF CHANGE** *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME *(Last, First, Middle Initial)* | b. ORGANIZATION | |
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